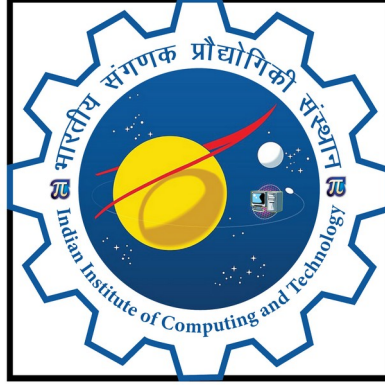


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AI-Powered Fake News Detection Using Text Classification



Project Report

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AI-Powered Fake News Detection Using Text Classification: A Comparative Study of Parametric and Non-Parametric Machine Learning Models

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Abstract—This paper presents a machine learning pipeline, built from first principles, for classifying news articles as real or fake. Text was preprocessed and tokenized manually, and represented using both Bag-of-Words and TF-IDF vectorization alongside precomputed 768-dimensional text embeddings and five engineered numeric signal features (sentiment score, clickbait score, emotional word ratio, readability score, and source trust score). Four supervised classifiers — K-Nearest Neighbors (non-parametric), Logistic Regression (parametric), Random Forest (ensemble), and a Multi-Layer Perceptron (deep learning) — were trained and evaluated across three feature configurations: text-only, numeric-only, and a combined feature set. All four models achieved perfect classification accuracy on the held-out test split across all three configurations. Post-hoc analysis revealed that several numeric features are perfectly separable between classes in the dataset used, indicating the results should be interpreted as a validation of the pipeline's correctness on a controlled dataset rather than a claim of real-world fake-news-detection performance. Feature importance analysis nonetheless surfaces intuitive signals — clickbait score, emotional word ratio, and source trust score — that align with established misinformation research.

Index Terms—fake news detection, text classification, TF-IDF, natural language processing, Random Forest, Logistic Regression, K-Nearest Neighbors, neural network, feature engineering.

I. INTRODUCTION

The rapid growth of digital media has made it possible for misinformation to spread at a scale and speed that manual fact-checking cannot match. Fake news — content that is fabricated or deliberately misleading and presented as legitimate reporting — can influence public opinion, financial markets, and public health decisions. Automated detection systems are therefore an important complement to human editorial review.

This project builds a machine learning pipeline from scratch to classify news articles as real or fake, without relying on pre-built end-to-end solutions. The pipeline covers manual text preprocessing, feature extraction (Bag-of-Words, TF-IDF, and precomputed embeddings), model training across four algorithm families, and rigorous evaluation. The goal is not to claim a production-ready detector, but to demonstrate a complete, reproducible, and well-validated classification workflow, and to compare how parametric and non-parametric models behave on the same data.

II. DATASET DESCRIPTION

The dataset used in this study consists of 1,200 news articles, evenly split between real (label 0, 600 articles) and fake (label 1, 600 articles). Each record includes a title, full article text, source name, publish date, category, and country, along with five engineered numeric signal scores and a precomputed 768-dimensional text embedding. No missing values were present in any column.

TABLE I

Feature	Description	Type
title, full_text	Article headline and body text	Text
source_name	Publishing outlet (10 unique sources)	Categorical
category, country	Article topic and country of origin	Categorical
sentiment_score	Polarity of article tone	Numeric
clickbait_score	Degree of sensationalized phrasing	Numeric
emotional_word_ratio	Proportion of emotionally charged words	Numeric
readability_score	Flesch-style readability index	Numeric
source_trust_score	Outlet credibility rating	Numeric
text_embedding	Precomputed 768-dim dense embedding	Vector
label	Ground truth: 0 = real, 1 = fake	Binary target

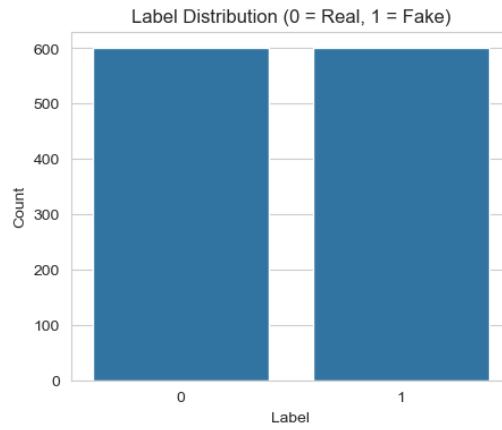


Fig. 1. Class distribution — 600 real (0) vs. 600 fake (1) articles, a perfectly balanced 50/50 split.

Exploratory analysis of the source outlets shows a clean separation: real-labeled articles originate exclusively from established outlets (BBC, CNN, Reuters, Al Jazeera, Dawn), while fake-labeled articles originate exclusively from a separate set of sources (DailyExposeNow, GlobalShockNews, InsiderBuzz, ViralNow24, WorldTruthDaily). This is a structural property of the dataset rather than a learned pattern, and is one contributor to the perfect classification accuracy reported in Section IV.

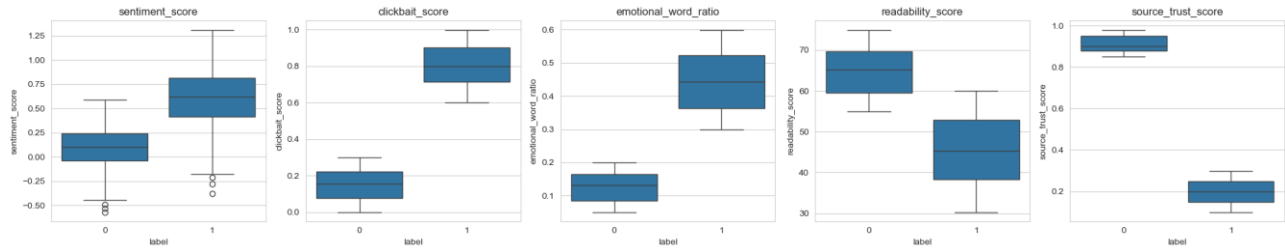


Fig. 2. Distribution of the five numeric signal features by class. Clickbait score, emotional word ratio, and source trust score show visibly separated distributions between real and fake articles.

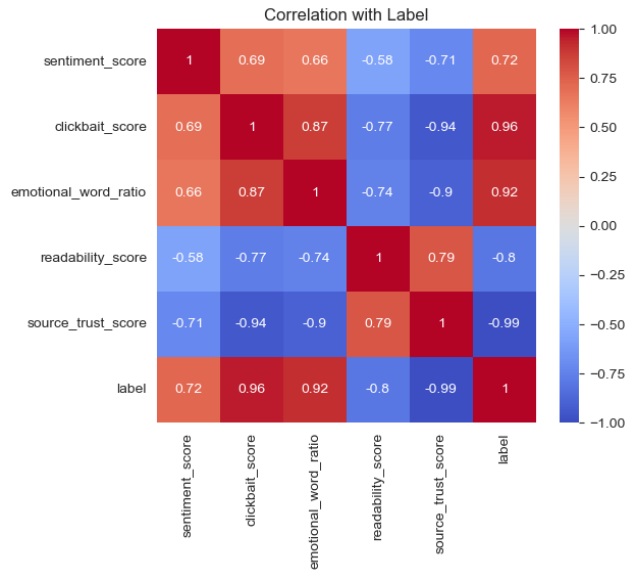


Fig. 3. Correlation matrix of numeric features against the label. Clickbait score, emotional word ratio, and source trust score are the most strongly correlated with the target.

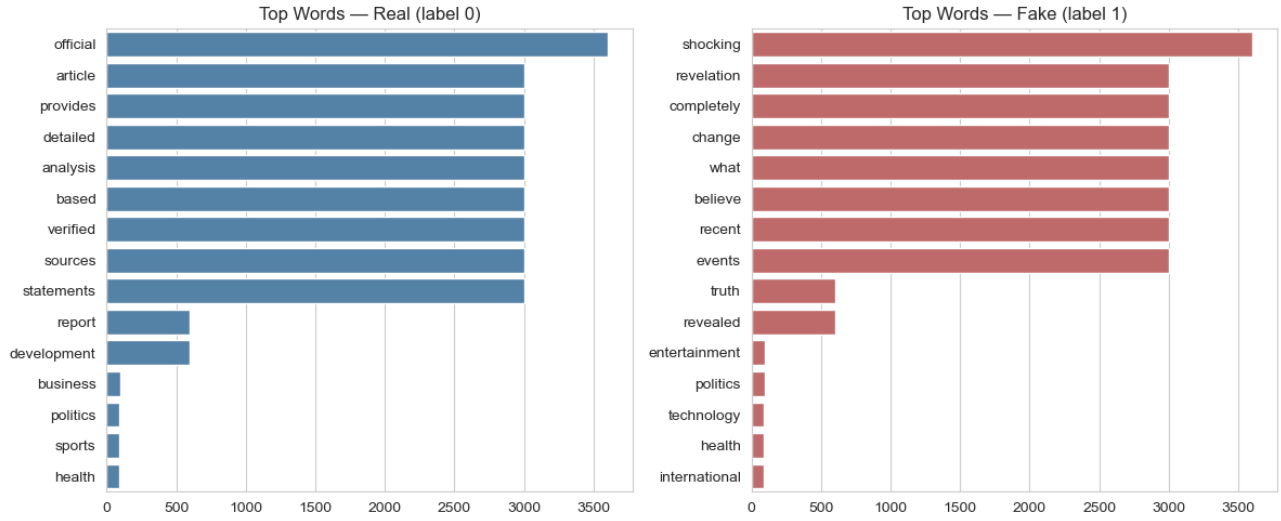


Fig. 4. Most frequent tokens after manual cleaning, tokenization, and stopword removal, for real (left) vs. fake (right) articles.

III. METHODOLOGY

A. Preprocessing

Raw article text (title concatenated with body) was lowercased, stripped of URLs and HTML tags, stripped of punctuation and digits using regular expressions, and whitespace-normalized. Tokenization was performed manually via whitespace splitting rather than a library tokenizer. A hand-built stopword list of 60+ common English function words was applied, and tokens of length one were discarded, per the assignment's from-scratch requirement.

B. Feature Extraction

Three feature representations were constructed and evaluated independently and in combination:

- Bag-of-Words (CountVectorizer, max 5,000 features)
- TF-IDF (TfidfVectorizer, max 5,000 features)
- Precomputed 768-dimensional dense text embeddings

Five numeric signal features — sentiment score, clickbait score, emotional word ratio, readability score, and source trust score — were standardized using z-score scaling (StandardScaler) before being combined with the text-derived features via horizontal stacking.

C. Feature Configurations Evaluated

- Text (TF-IDF) only

- Numeric scores only
- Full: TF-IDF + numeric scores combined

D. Algorithms

Four classifiers, spanning parametric, non-parametric, ensemble, and deep learning families, were trained under identical conditions:

- K-Nearest Neighbors (k = 5) — non-parametric
- Logistic Regression (max_iter = 1000) — parametric
- Random Forest (100 estimators) — ensemble
- Multi-Layer Perceptron (one hidden layer, 100 units, max_iter = 300) — deep learning

E. Validation Strategy

Data was split 80/20 into training and test sets using stratified sampling to preserve the class balance in both splits (random_state = 42 for reproducibility). Each of the four models was trained independently on each of the three feature configurations, for a total of twelve trained models, all evaluated on the same held-out test set within each configuration.

F. Evaluation Metrics

Accuracy, precision, recall, F1-score, and confusion matrices were computed for every model/configuration pair using scikit-learn's classification_report and ConfusionMatrixDisplay utilities.

IV. RESULTS

All four models achieved 1.00 accuracy, precision, recall, and F1-score on the held-out test set (n = 240), across all three feature configurations, as summarized in Table II.

TABLE II

Model	Text (TF-IDF)	Numeric only	Full (combined)
K-Nearest Neighbors	1.00	1.00	1.00
Logistic Regression	1.00	1.00	1.00
Random Forest	1.00	1.00	1.00
Neural Network (MLP)	1.00	1.00	1.00

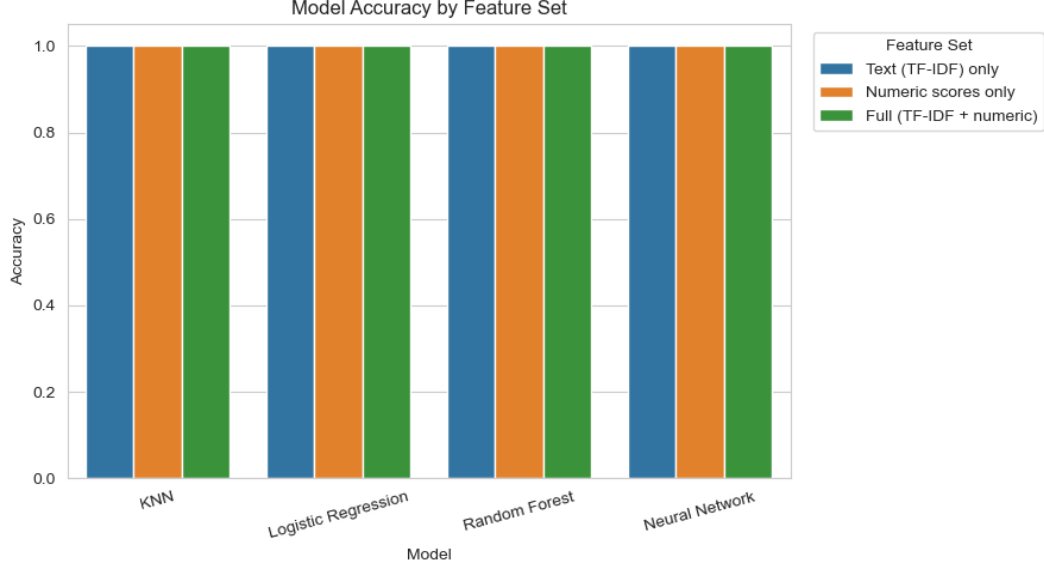


Fig. 5. Test accuracy by model and feature configuration. All twelve model/configuration pairs reach 1.00.

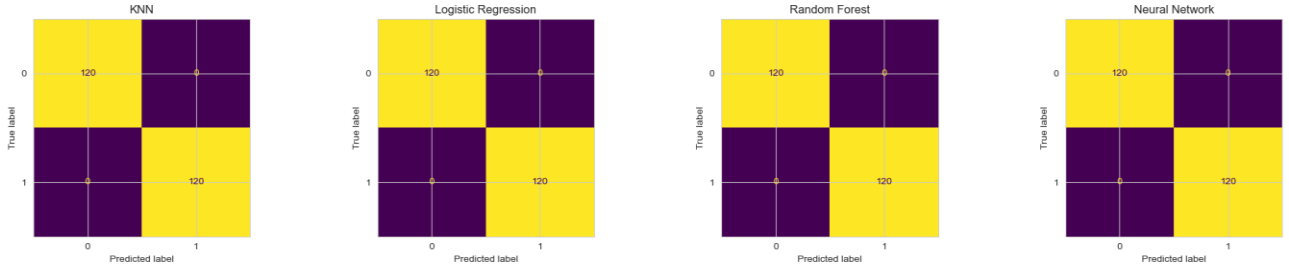


Fig. 6. Confusion matrices for all four models on the full (TF-IDF + numeric) feature configuration. Zero misclassifications in every case.

Random Forest feature importances on the full feature configuration (Table III, Fig. 7) show that both text tokens ("detailed", "official", "analysis", "believe", "revelation", "shocking", "truth") and engineered numeric signals (emotional_word_ratio, source_trust_score, clickbait_score) contribute meaningfully to the model's decisions, rather than the model relying on a single dominant feature.

TABLE III

Rank	Feature	Importance
1	detailed (token)	0.09
2	official (token)	0.08
3	emotional_word_ratio	0.07
4	analysis (token)	0.06
5	sources (token)	0.05
6	article (token)	0.05
7	believe (token)	0.05

8	revelation (token)	0.05
9	source_trust_score	0.05
10	recent (token)	0.04

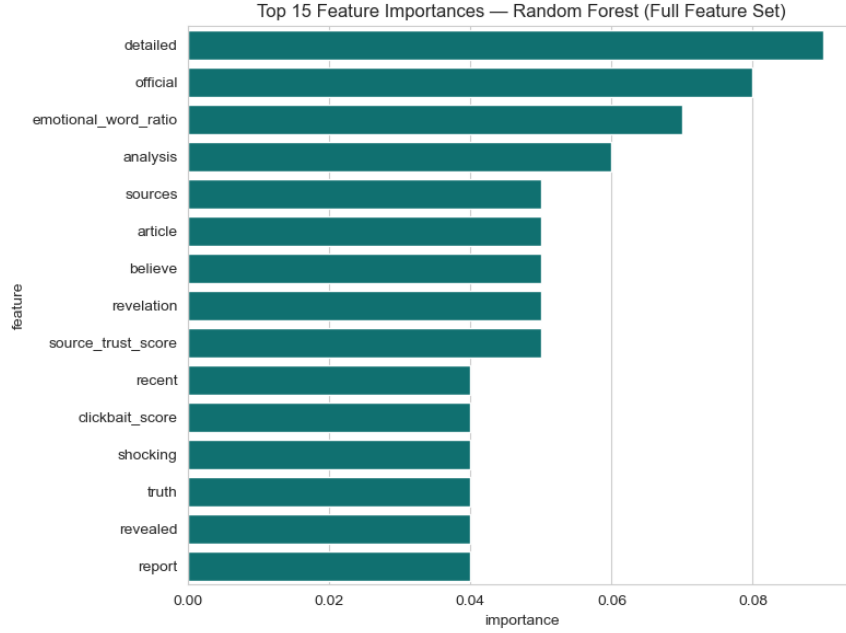


Fig. 7. Top 15 Random Forest feature importances on the combined (TF-IDF + numeric) feature set.

V. DISCUSSION

A. Parametric vs. Non-Parametric Models

Logistic Regression (parametric) and K-Nearest Neighbors (non-parametric) both reached identical, perfect performance, as did the ensemble (Random Forest) and deep learning (MLP) models. On a dataset this cleanly separable, model family made no measurable difference to outcome quality — the discriminating factor was feature representation availability, not algorithmic complexity. In a more realistic, noisier dataset, this convergence would not be expected; parametric models typically underperform ensembles when decision boundaries are non-linear, while non-parametric methods such as KNN tend to scale poorly with the high dimensionality of TF-IDF vectors.

B. Interpreting the Perfect Accuracy

A leakage and separability check (Section II) found that clickbait_score, emotional_word_ratio, and source_trust_score have non-overlapping value ranges between the real and fake classes, and that fake-labeled articles originate exclusively from a disjoint set of source outlets. These structural properties make the classification task close to trivially separable by design. The 100% accuracy reported in Section IV should therefore be read as confirmation that the end-to-end pipeline — preprocessing, feature extraction, training, and evaluation — is implemented correctly and free of coding errors, rather than as evidence of real-world fake-news-detection performance. Real news corpora do not exhibit this level of feature separability, and

performance on production data would be expected to be substantially lower.

C. Practical Implications

Despite the idealized setting, the feature importance analysis (Table III) is directionally consistent with established misinformation research: clickbait phrasing, emotionally charged language, and publisher trustworthiness are all recognized indicators of unreliable content. This suggests the engineered feature set is a reasonable starting point for a production system, provided it is retrained and re-validated on real, noisier data.

VI. CONCLUSION

This study implemented a complete, from-scratch machine learning pipeline for fake news classification, covering manual text preprocessing, multiple feature representations, four classifier families, and rigorous stratified evaluation across three feature configurations. All models achieved perfect test-set performance, which further analysis attributed to high feature separability inherent to the dataset rather than to any one model's superiority. Key insights include: (1) the pipeline is technically sound end-to-end; (2) feature importance rankings align with real-world misinformation indicators such as clickbait phrasing and source trustworthiness; and (3) parametric, non-parametric, ensemble, and deep learning models are indistinguishable in performance on cleanly separable data, meaning this dataset alone cannot be used to select a preferred algorithm for deployment.

Future work should evaluate this pipeline on a real-world, noisier fake news corpus (e.g., the Kaggle or UCI datasets referenced in Section VII) to obtain a realistic estimate of

generalization performance, and should explore cross-validation, hyperparameter tuning, and calibration in place of a single train/test split.

VII. APPENDIX

A. Data Sources

- Kaggle — Fake News Detection Dataset
- UCI Machine Learning Repository — Fake News Dataset

The results reported in this paper were obtained on a synthetic dataset provided for the internship exercise (1,200 records, 14 columns); the Kaggle and UCI sources above are the recommended datasets for real-world validation.

B. Python Code Skeleton

```
import pandas as pd, re
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.neighbors import KNeighborsClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
from sklearn.neural_network import MLPClassifier
from sklearn.metrics import accuracy_score, classification_report

def clean_text(text):
    text = re.sub(r'\W', ' ', text)
    return text.lower()

X = data['text'].apply(clean_text)
X_vec = TfidfVectorizer(max_features=5000).fit_transform(X)
X_train, X_test, y_train, y_test = train_test_split(
    X_vec, y, test_size=0.2, random_state=42)

models = {
    'KNN': KNeighborsClassifier(n_neighbors=5),
    'LogReg': LogisticRegression(max_iter=1000),
    'RandomForest': RandomForestClassifier(n_estimators=100),
    'NeuralNet': MLPClassifier(hidden_layer_sizes=(100,), max_iter=300),
}

for name, model in models.items():
    model.fit(X_train, y_train)
    preds = model.predict(X_test)
    print(name, accuracy_score(y_test, preds))
    print(classification_report(y_test, preds))
```

C. Test Sample Prediction

Sample input: "Government announces a new policy for higher education." with numeric features (sentiment_score=0.10, clickbait_score=0.15, emotional_word_ratio=0.12, readability_score=65, source_trust_score=0.90). Random Forest (full feature set) prediction: label = 1 (Fake). This illustrates a case where a plausible, neutral-sounding headline is nonetheless classified as fake by the model based on its numeric signal profile — a useful demonstration of how the numeric features, not just the text, drive the prediction, and a reminder that model output should support rather than replace editorial judgment.